WORKED ANSWERS

Graphs

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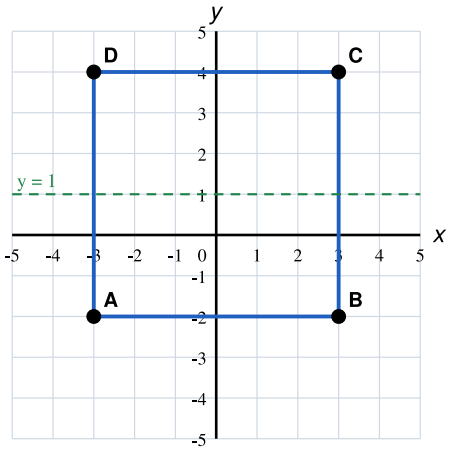
Key ideas: For $y = mx + c$, the gradient is m and the intercept is c. Lines with equal gradients are parallel. If $m > 0$ the line makes an acute angle with the positive x-axis; if $m < 0$ the angle is obtuse.

Question 1

(i), (ii) Side AB = 6 units, so C = **(3, 4)** and D = **(−3, 4)**.

(iii) AB: **y = −2** BC: **x = 3** CD: **y = 4** DA: **x = −3**

(iv) Axes of symmetry: **x = 0** (the y-axis) and **y = 1**.



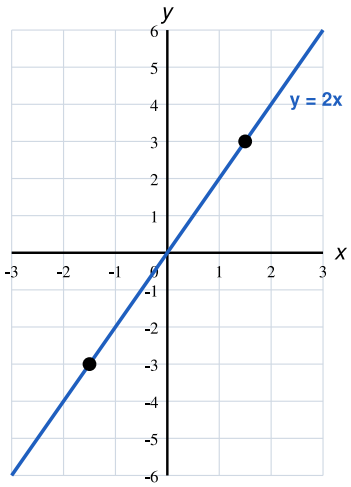
Question 2

x	−2	−1	0	1	2
y	−4	−2	0	2	4

(iii) When x = 1.5, **y = 3**

(iv) When y = −3, **x = −1.5**

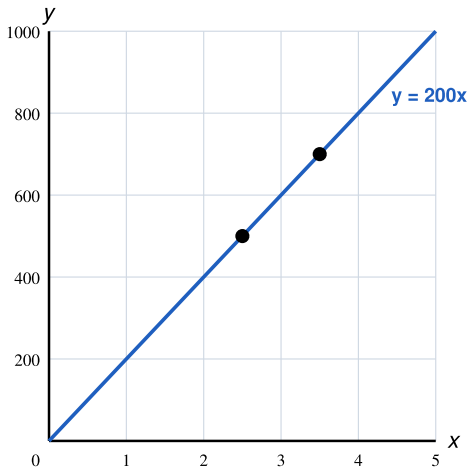
(v) **No.** When x = −1, y = 2 × (−1) = −2, not 2.



Question 3

(i) **y = 200x**

x (kg)	1	2	3	4
y (Rs)	200	400	600	800

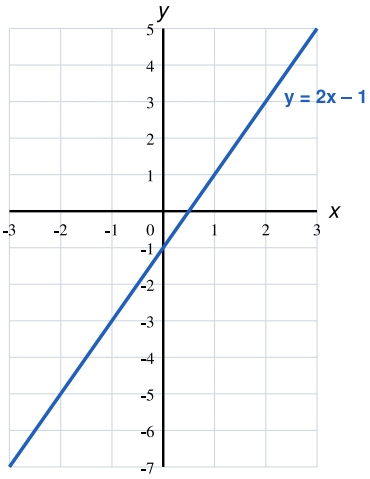


(iv) Price of 2.5 kg = **Rs 500**

(v) Rs 700 buys **3.5 kg**

Question 4

x	−2	−1	0	1	2
y	−5	−3	−1	1	3



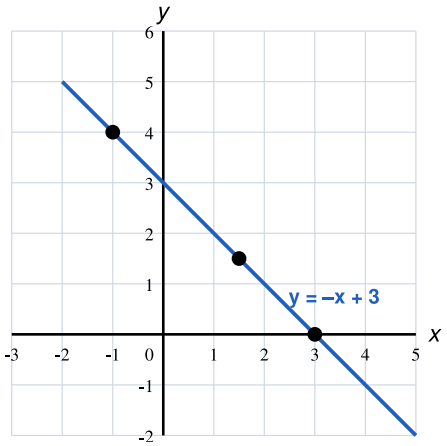
(iii) Gradient m = **2**, intercept c = **−1**

(iv) k = 2(4) − 1 = **7**

(v) Same gradient 2, intercept 2: **y = 2x + 2**

Question 5

x	−2	−1	0	1	2	3	4
y	5	4	3	2	1	0	−1



(iii) Intercept = **3**; meets the x-axis at **(3, 0)**

(iv) When x = 1.5, **y = 1.5**; when y = 4, **x = −1**

(v) **Obtuse**, because the gradient (−1) is negative.

Question 6

(i) (d) $2y = 4x + 6 \rightarrow \mathbf{y = 2x + 3}$ (e) $y + 3x = 1 \rightarrow \mathbf{y = -3x + 1}$

Line	(a)	(b)	(c)	(d)	(e)	(f)
Gradient m	3	−2	½	2	−3	3

Line	(a)	(b)	(c)	(d)	(e)	(f)
Intercept c	2	5	−4	3	1	−1

(iii) Acute (m positive): **(a), (c), (d), (f)** Obtuse (m negative): **(b), (e)**

(iv) **(a) and (f)** are parallel, because both have the same gradient, 3.

Question 7

(i) $\mathbf{y = 3x - 2}$

(ii) $\mathbf{y = -x + 4}$

(iii) Gradient 2 (parallel), $c = -3 \rightarrow \mathbf{y = 2x - 3}$

(iv) $c = 0$; $m = 6 \div 2 = 3 \rightarrow \mathbf{y = 3x}$

(v) $c = 1$; $m = (5 - 1) \div (2 - 0) = 2 \rightarrow \mathbf{y = 2x + 1}$

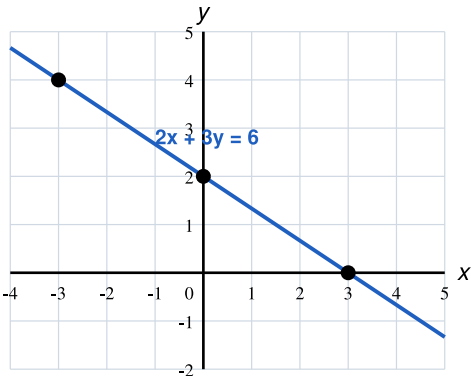
Question 8

(i) $3y = -2x + 6 \rightarrow \mathbf{y = -(2/3)x + 2}$

(ii) Gradient = **−2/3**, intercept = **2**

(iii) When $x = 0$, $y = 2 \rightarrow$ **(0, 2)**. When $y = 0$, $2x = 6$, $x = 3 \rightarrow$ **(3, 0)**

(v) $2(-3) + 3(4) = -6 + 12 = 6$, which equals the right-hand side, so (−3, 4) lies on the line.



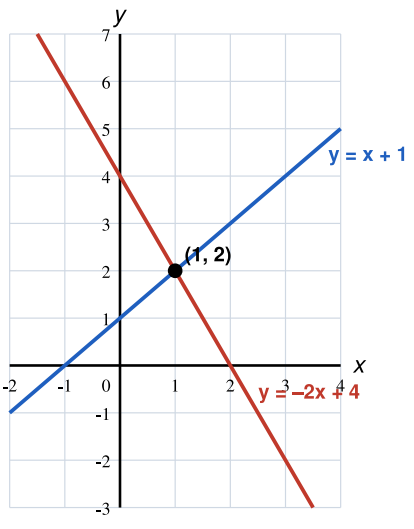
Question 9

y = x + 1

x	−1	0	1	2	3
y	0	1	2	3	4

y = −2x + 4

x	−1	0	1	2	3
y	6	4	2	0	−2



(iii) y = x + 1: m = **1**, c = **1**. y = −2x + 4: m = **−2**, c = **4**

(iv) Point of intersection = **(1, 2)**

(v) **x = 1**